

Gujarat Vidyapith		
Department of Computer Science		
Course : Msc(IT) Semester-I	Year : 2026-27	Subject Teacher : Prof. (Dr.) Ajay Parikh
Sub : Lab based on Python Programming and Application Development		Enrollment No.:_____

Unit-1 & Unit-2

Practical No.	Lab Activity	Date Assigned	Date Submission	Sign	Grade
1	Consecutive Duplicate Detector Accept N integers. Display only those numbers that appear consecutively more than once. Input: 1 2 2 3 4 4 4 5 Output: 2 4	23/7/2026	6/8/2026		
2	Missing Roll Number Roll numbers should be from 1 to N. One roll number is missing. Find the missing roll number without sorting. Example: 1 2 3 5 6 Output: 4	23/7/2026	6/8/2026		
3	Password Strength Analyzer Accept password. Check • uppercase • lowercase • digit • special character • repeated consecutive characters Display all failed rules.	30/7/2026	6/8/2026		
4	Write a Python program that accepts a paragraph from the user. The program should: • Split the paragraph into words and store them in a list. • Display: ○ Total number of words ○ Number of unique words ○ Longest word ○ Shortest word	6/8/2026	7/8/2026		

	○ Words appearing more than once				
5	Student Rank Processing Engine Develop a Python application that reads student records from User input. Each record contains: ● Roll No ● Name ● Marks in 5 subjects The program must: ● Compute total and percentage. ● Assign grades. ● Handle ties while assigning ranks (students with equal marks receive the same rank). ● Display the next rank correctly (e.g., 1,2,2,4).				
6	Student Rank Processing Engine using Functions and Modules Develop a Python application to process and rank student records stored in an Excel file. Organize the application using user-defined modules and functions. Input The input Excel file contains the following fields: ● Roll No ● Name ● Marks in Subject 1 ● Marks in Subject 2 ● Marks in Subject 3 ● Marks in Subject 4 ● Marks in Subject 5 Create the following modules: student.py Implement functions to: ● Read student records from an Excel file. ● Calculate the total marks and percentage for each student. ● Assign grades based on the percentage. ● Return the processed student records. ranking.py Implement functions to: ● Rank students based on total marks. ● Handle tied ranks correctly (e.g., 1, 2, 2, 4). ● Sort students in descending order of marks. ● Return the ranked student records. report.py Implement functions to: ● Display the ranked student records in a formatted table. ● Export the ranked results to a new Excel file. main.py				

	The main program should: 1. Import the required functions from all modules. 2. Read student records from the input Excel file. 3. Calculate total marks, percentage, and grade for each student. 4. Generate student ranks while correctly handling ties. 5. Display the ranked result in rank order. 6. Export the final ranked result to another Excel file. 7. Handle file-related errors and invalid data using exception handling.				
7	Course Registration Conflict Detector using Object-Oriented Programming Develop a Python application to help a college identify timetable conflicts when students register for multiple courses. Create the following classes: • Course ○ Course Code ○ Day of the Week ○ Start Time ○ End Time • Student ○ Student ID ○ Student Name ○ List of Registered Courses Implement appropriate constructors (<code>__init__()</code>) to initialize all objects. The application should: 1. Accept details of N students. 2. For each student, accept details of multiple registered courses. 3. Detect whether any registered courses overlap on the same day. 4. Display all conflicting course pairs. 5. If no conflict exists, display the student's timetable. 6. Calculate and display the total number of registered courses for each student. 7. Override the <code>__str__()</code> method to display student and course information in a readable format. 8. Handle invalid time entries or incorrect input using exception handling.				
8	Develop a Flask web application called AI Prompt Management Portal that allows users to organize and search prompts used for AI applications.				
9	Extend the AI Prompt Management Portal developed previously by integrating database connectivity, session management, file handling, and RESTful APIs.				

Grade	Marks
A	18 - 20
B	14 - 17
C	11 - 13
D	8 - 10
E	5 - 7
F	1 - 4
S	Absent